Week 02 Progress Report

Details

- **Name of Student:** Tirthak Girish Likhar
- **PRN: 22070521041**
- **Name of Industry:** SS INFOTECH
- **Reporting Week: (12/01/2026 to 16/01/2026)**
- **Institute Mentor Name:** Dr. Shreyas Hole
- **Industry Mentor Name:** Mr. Viraj Patle

Overview of a Week

- In Week 2 of my internship, I focused on understanding basic networking concepts and how network-related activities are logged inside a Linux system. My industry mentor assigned me hands-on tasks to help me get comfortable with identifying network interfaces, checking connectivity, and reading different types of system logs.

- I started by learning how to view my Ubuntu virtual machine's network configuration using commands like `ip a`, `ip route`, and `ping`. This helped me understand how my system receives an IP address and connects to the network. I also checked active ports and services using commands such as `ss -tulpn` and `netstat`, which gave me a clear idea of which applications are listening for network connections.

- After that, I spent time exploring important Linux log files like `syslog`, `auth.log`, and Network Manager logs. These logs are very important for SOC work because they record system events, authentication attempts, and network changes. Going through them helped me understand what normal system behaviour looks like, which will later help me detect unusual or suspicious activity when I start working with the SIEM and incident simulation tasks.

- Overall, Week 2 helped me build a strong foundation in networking from a SOC perspective and improved my confidence in reading and interpreting Linux logs.

Key Tasks

- Checked the network configuration of the Ubuntu 24.04 VM using commands like ip a, ip route, and ping to understand IP assignment and connectivity.
- Identified active ports and listening services using ss -tulpn, ss -tunap, and netstat -tulnp.
- Explored important Linux log files such as /var/log/syslog and /var/log/auth.log to understand system events and authentication activities.
- Used log filtering techniques (e.g., grep) to check for SSH-related entries in the authentication logs.
- Reviewed NetworkManager logs using journalctl to analyze network interface state changes and DHCP events.
- Captured and organized screenshots of all commands and logs for project documentation and weekly reporting.

Key Accomplishments

- Successfully analyzed the network configuration of the Ubuntu 24.04 virtual machine and understood how IP addresses and gateways are assigned in a virtualized environment.
- Identified and reviewed active ports, listening services, and network connections to understand baseline network behavior of a Linux system.
- Gained hands-on experience in reading and interpreting critical Linux log files such as syslog, auth.log, and NetworkManager logs from a SOC perspective.
- Understood how authentication activities, sudo commands, and system events are recorded and monitored through logs.
- Developed the ability to differentiate between normal system behavior and potentially suspicious network or authentication activity.
- Strengthened practical skills in Linux networking and log analysis, which are essential for future SIEM integration and SOC incident analysis tasks.

Screenshots of Activities (if Any)

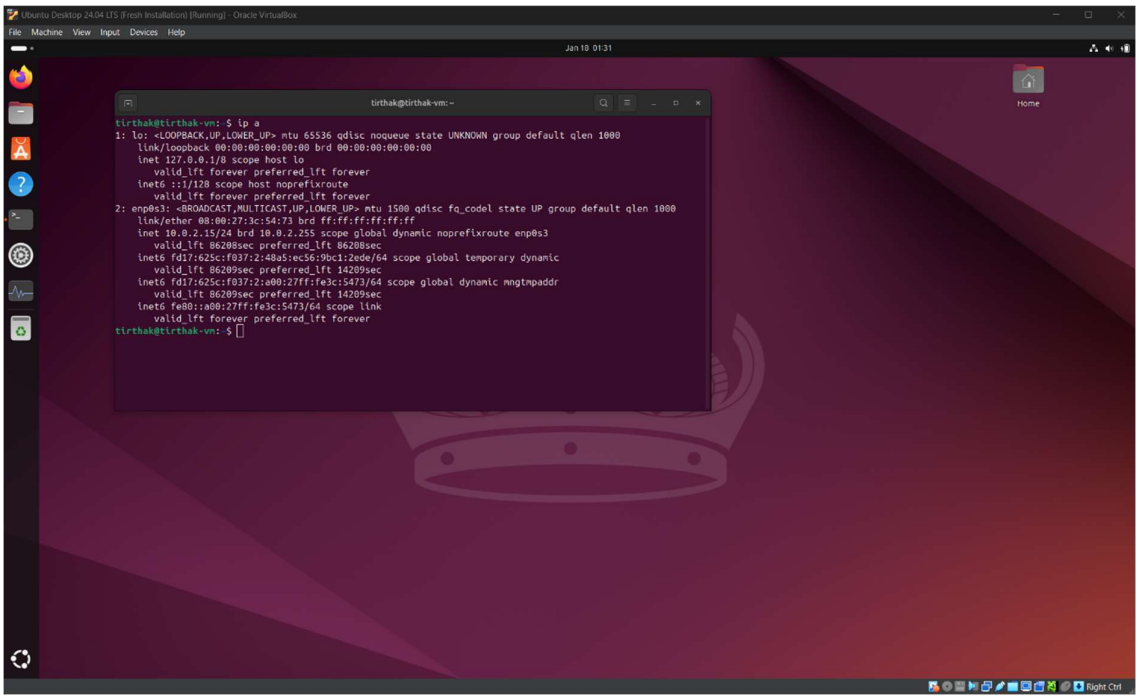


Fig: Output of ip a showing VM network interface and assigned IP address.

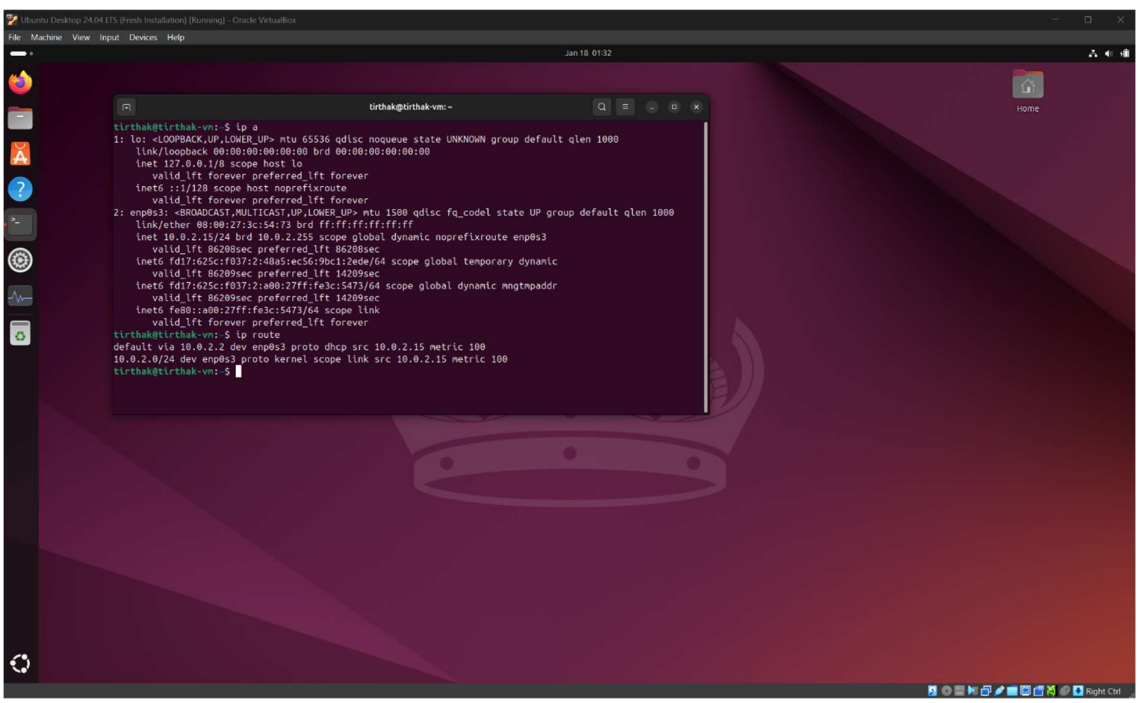
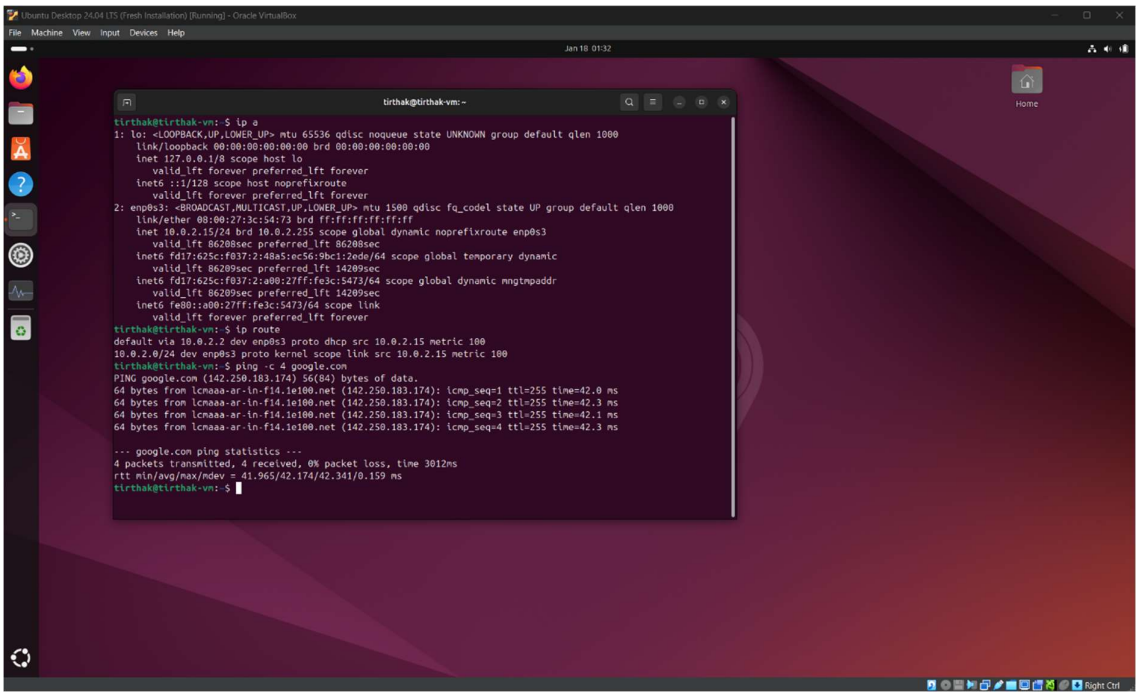


Fig: Output of ip route displaying default gateway and routing information.



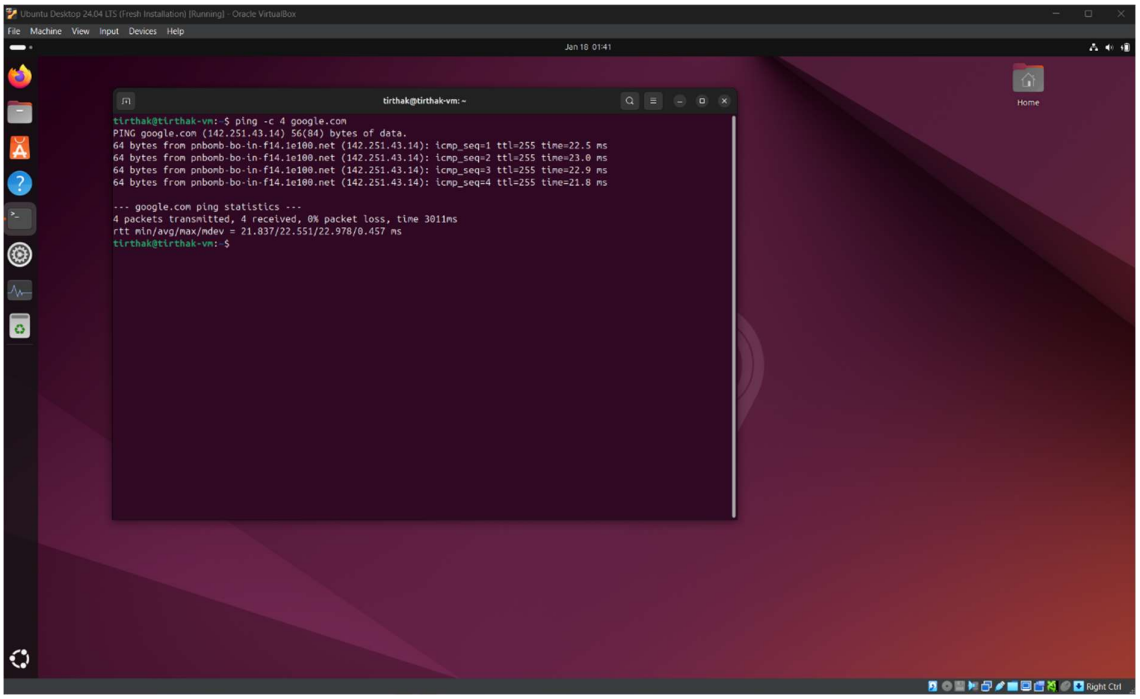


Fig: Successful connectivity test using ping command.

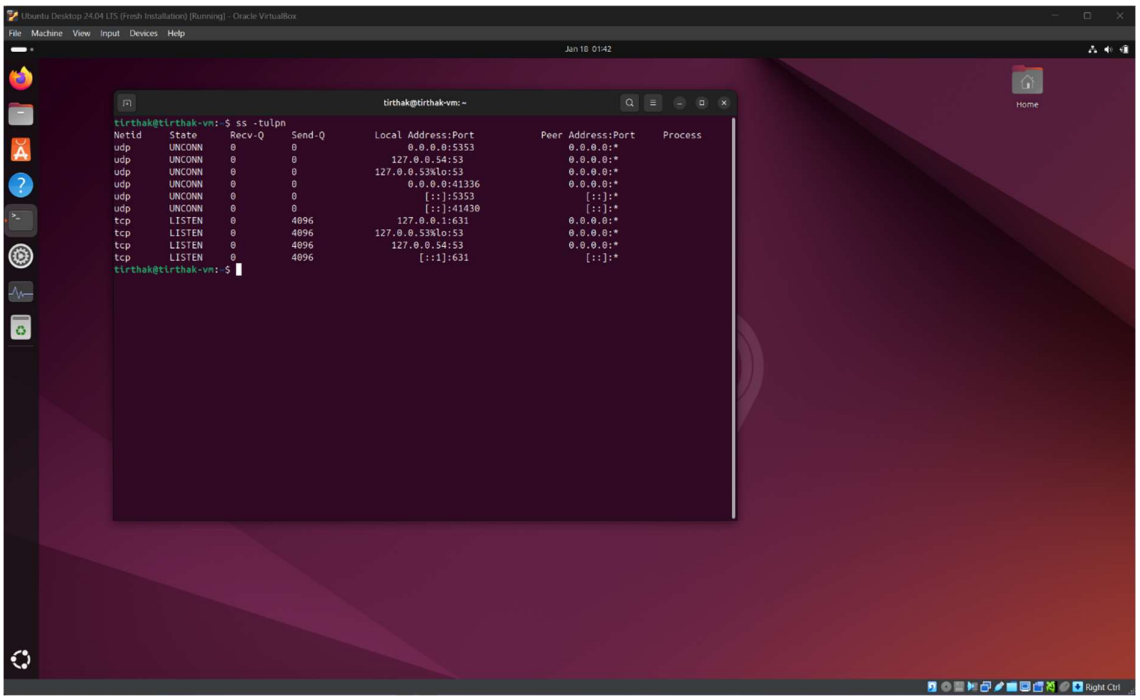


Fig: List of listening ports and services using ss -tulpn.

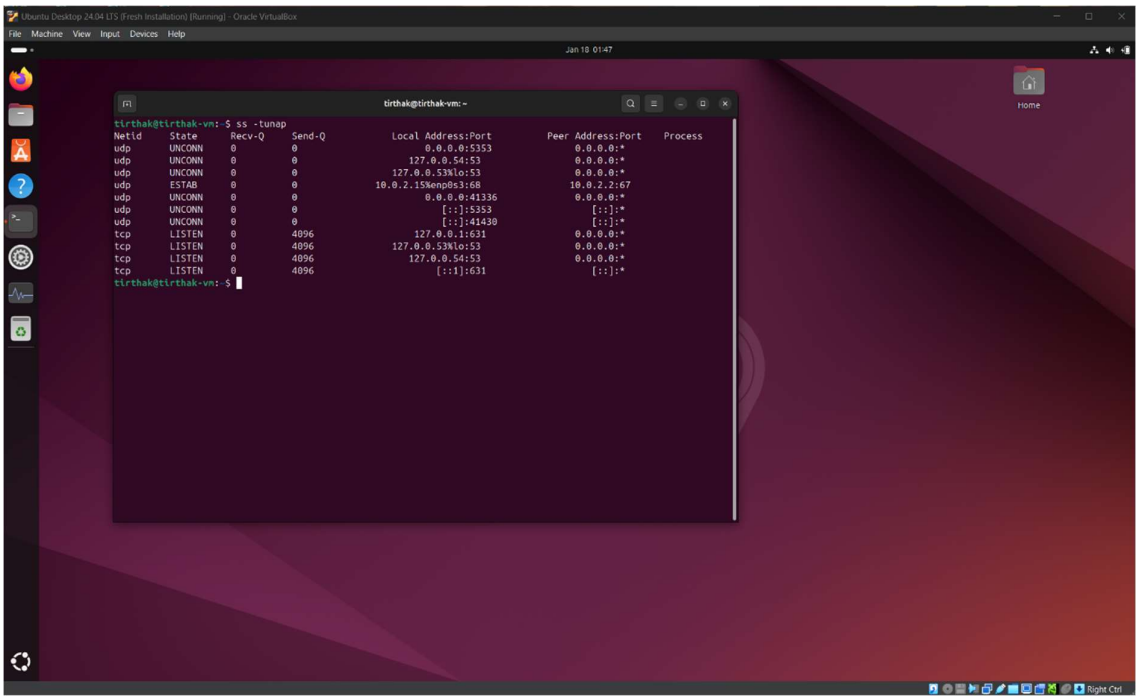


Fig: Active network connections displayed using ss -tunap.

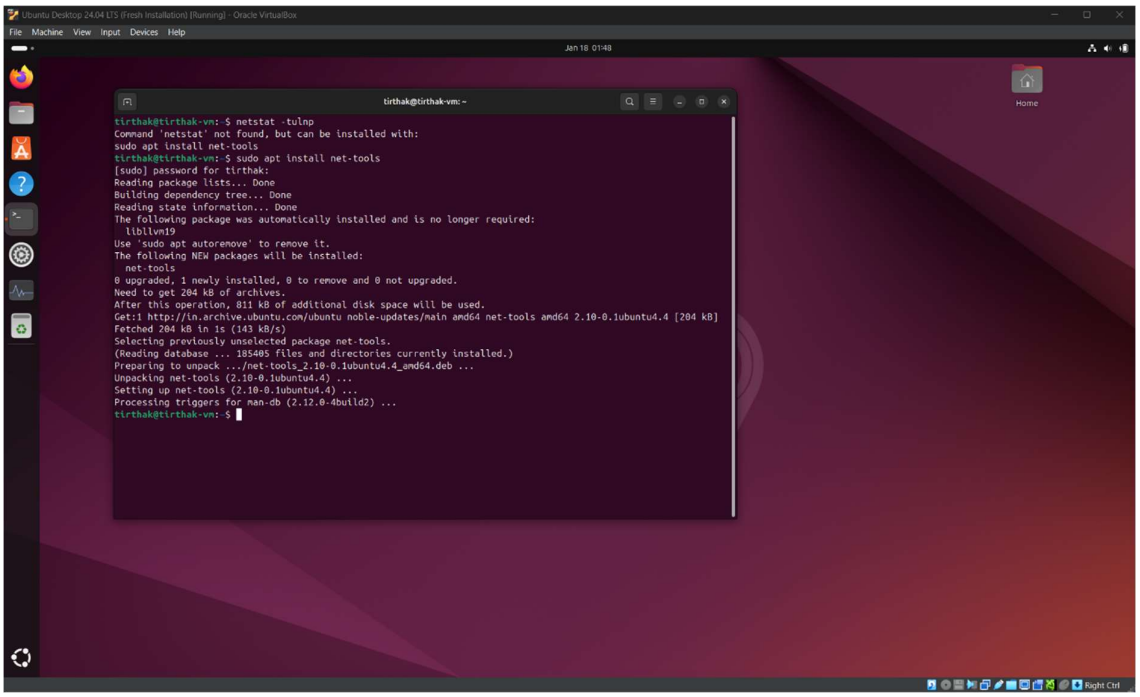


Fig: Downloading net-tools to execute the netstat command

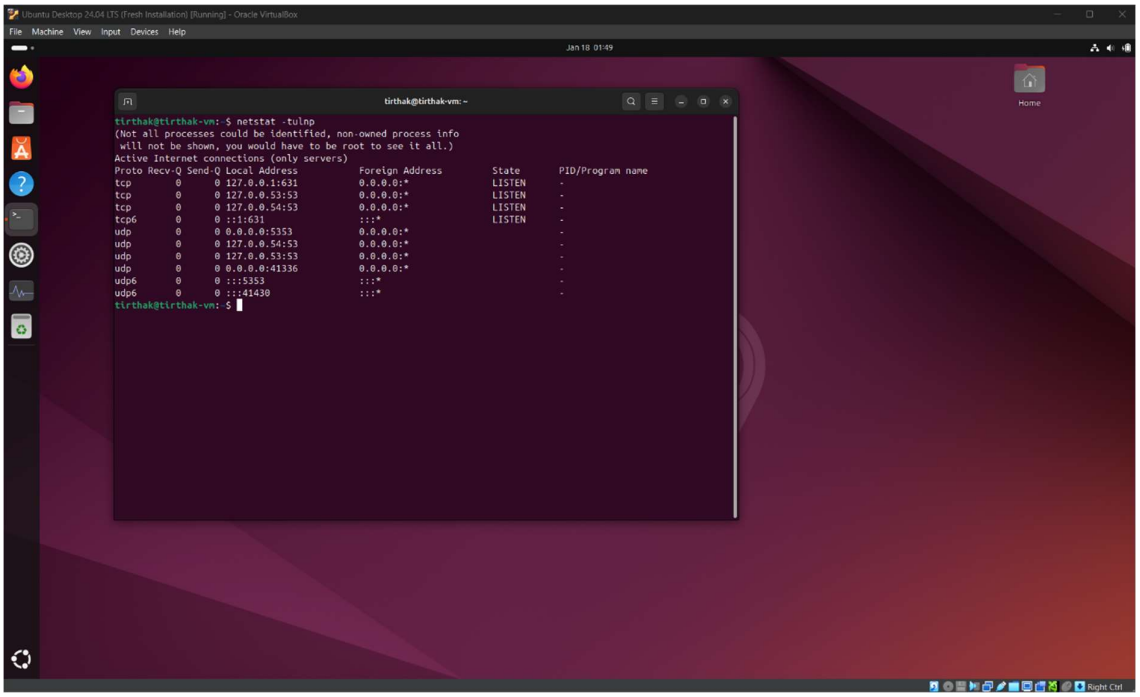


Fig: List of open ports using netstat -tulnp.

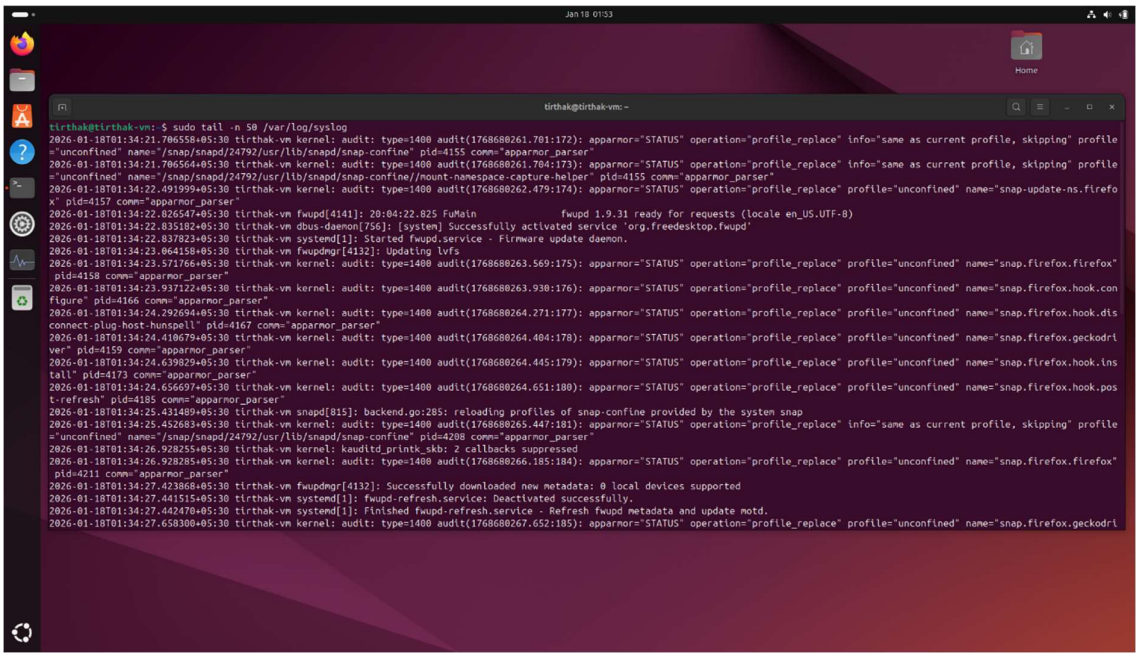


Fig: Recent system events from /var/log/syslog (system logs).

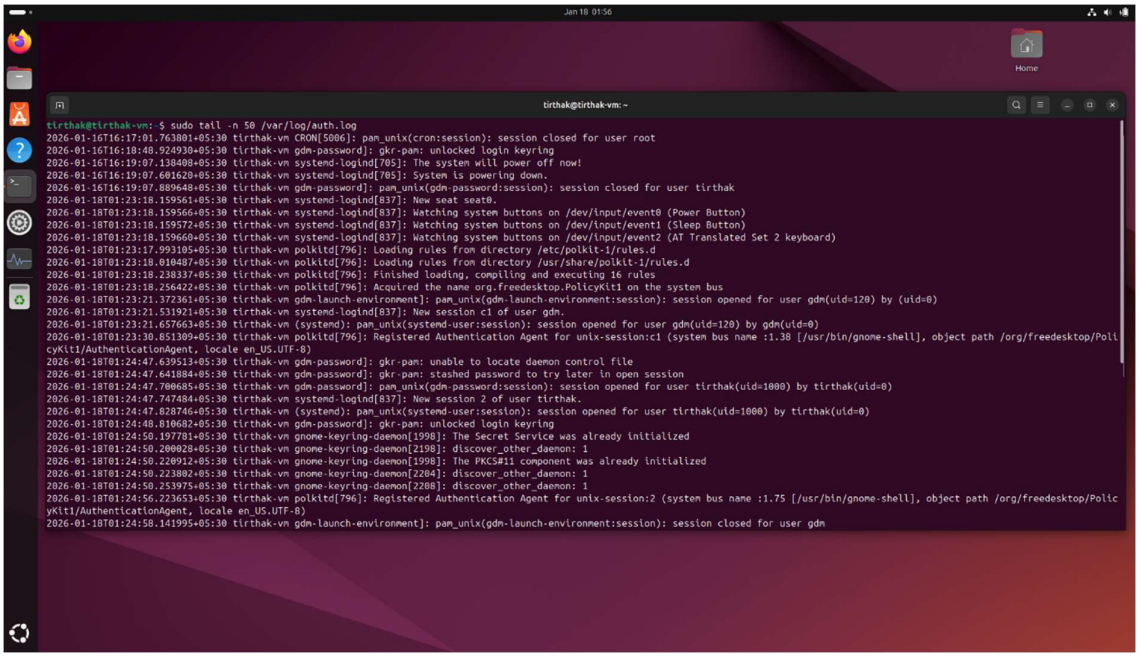


Fig: Authentication-related events from /var/log/auth.log.

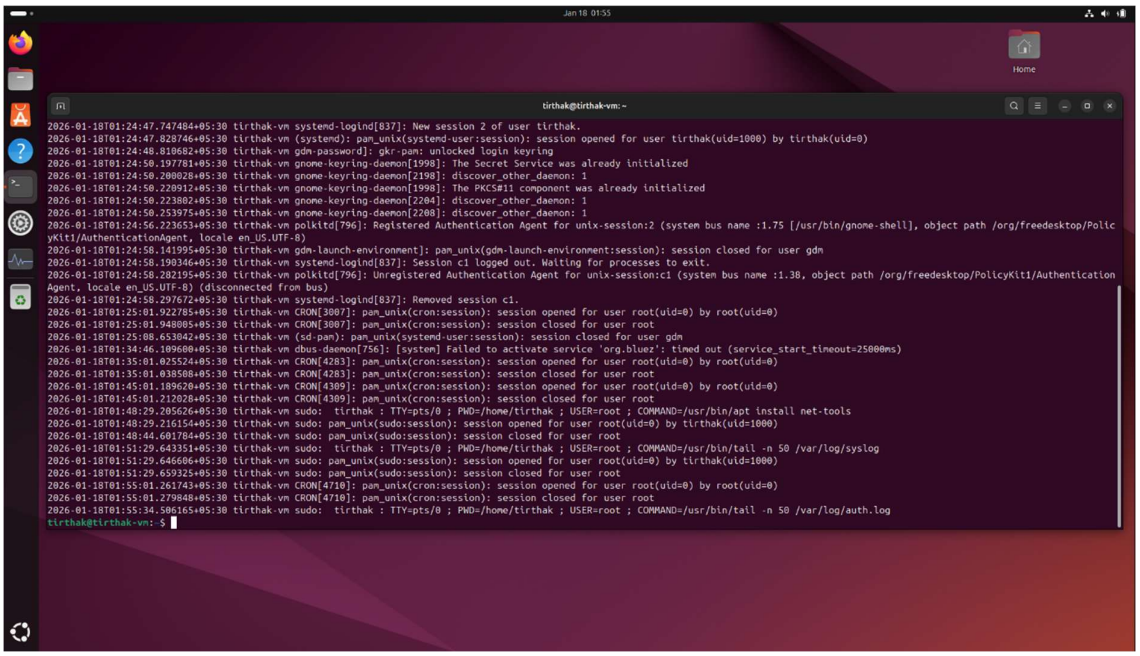


Fig: Authentication-related events from /var/log/auth.log.

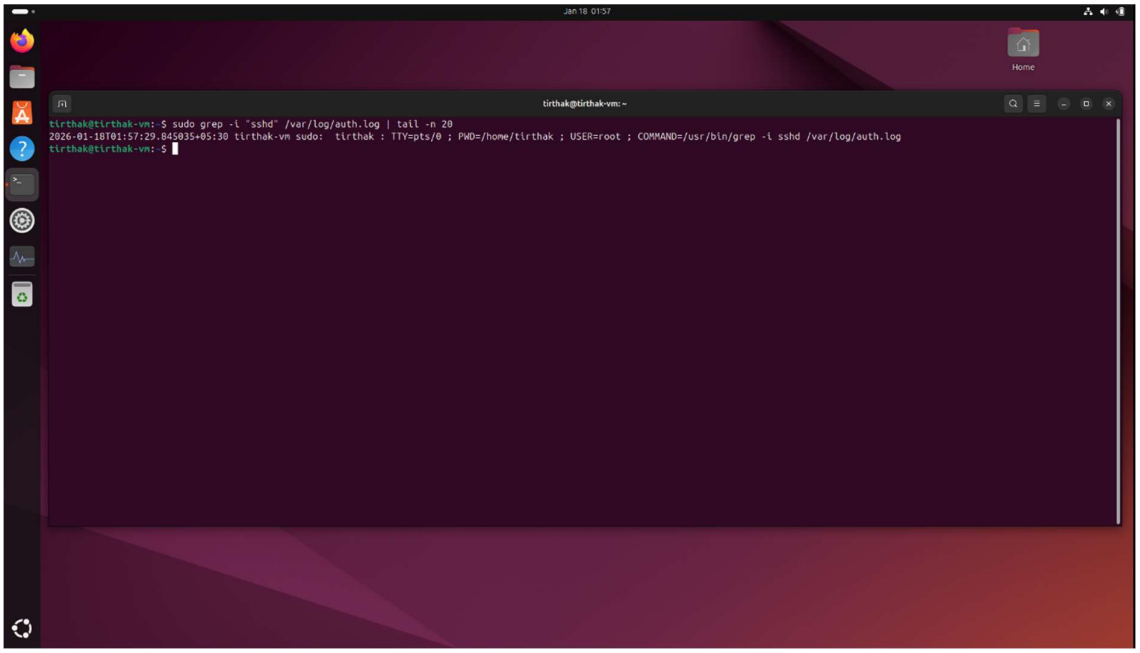


Fig: SSH-related log entries filtered using grep command.

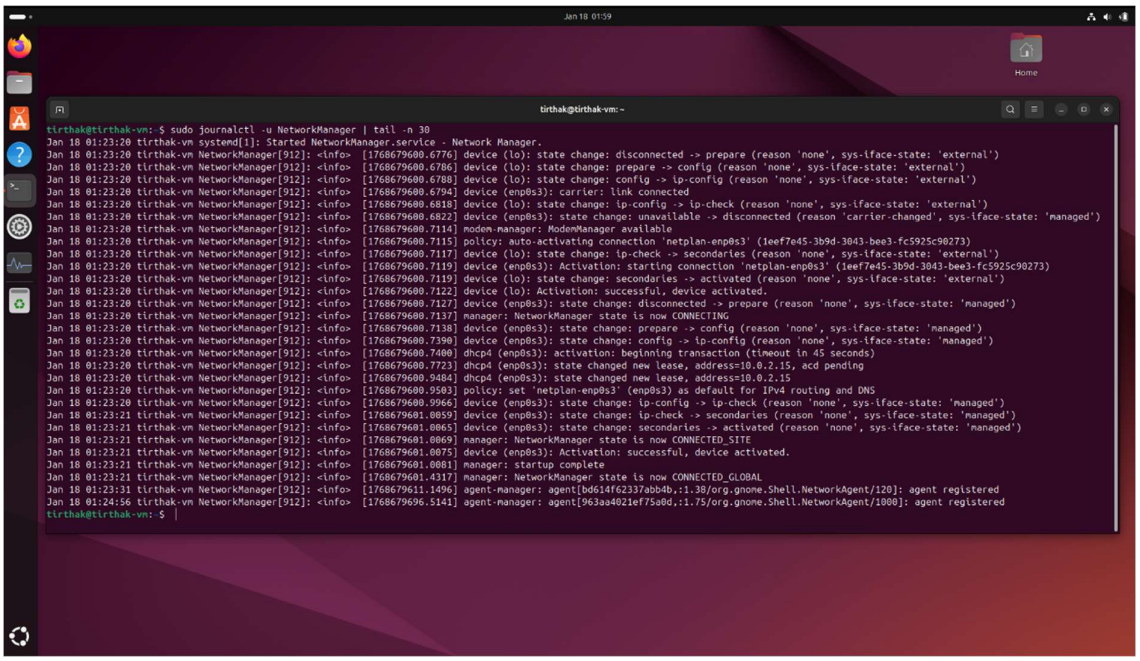


Fig: Network interface and DHCP events from NetworkManager logs.